22 Safe Use and Application of Electrical Apparatus

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Topic Highlights

Philosophy of General Purpose Requirements
Equipment for Use Where Explosive Concentrations of Gas, Vapor, or Dust Might be Present
Equipment for Use Where Combustible Dust May Be Present

22.1 Introduction

This article discusses ways to ensure electrical equipment does not endanger personnel or plant. See another chapter for a discussion of Process Safety and Safety Instrumented Systems—protecting the plant against the risk of equipment failing to perform its function in a control system or in a safety instrumented loop (SIL).

Developments during the past half century have greatly eased the selection of safe equipment. Standardization of general purpose safety requirements has made it possible to design a single product that is acceptable with very minor modifications, if any, in all nations. Worldwide adoption of common standards progresses for constructing and selecting equipment for use in hazardous locations, but transition from long-accepted national practices to adopt a somewhat different international or harmonized practice is, of necessity, slowed by historical differences in philosophy. Emphasis in this article is on the common aspects of design and use. Present day differences in national standards, codes and practices will be reduced in coming years. To ensure safety, a user anywhere must select, install, and use equipment in accordance with local standards and codes.

General purpose safety standards address construction requirements that ensure personnel will not be injured by electrical shock, hot surfaces or moving parts—and that the equipment will not become a fire hazard. Requirements for constructing apparatus to ensure it does not become a source of ignition of flammable gases, vapors, or dusts are superimposed on the general purpose requirements in equipment to be used where potentially explosive atmospheres may be present. The practice in all industrialized countries, and in many developing countries, is that all electrical equipment must be certified as meeting these safety standards. An independent laboratory is mandated for explosion-protected apparatus but, in some cases, adherence to general purpose requirements may be claimed by the manufacturer, subject to strict oversight by a third party. Thus, once a user has selected the equipment he or she wants to use, the user can be assured it meets the applicable construction standards if it is certified or approved. It is the user's duty to install and use the equipment in a manner that ensures that safety designed into the equipment is not compromised in use.

22.2 Philosophy of General Purpose Requirements

Protection against electrical shock is provided by construction rules that recognize voltages below about 30V alternating current (AC), 42.4V peak, or 60 volts direct current (DC) don't pose danger of electrocution in normal industrial or domestic use, whereas contact with higher voltages may be life threatening. Design rules, therefore, specify insulation, minimum spacings, or partitions between low voltage circuits and higher voltage ones to prevent them from contacting each other so that an accessible extra low voltage circuit becomes hazardous. Construction must ensure higher voltage circuits cannot be touched in normal operation or by accidental exposure of live parts. Any exposed metallic parts must be grounded or otherwise protected from being energized from hazardous voltages.

Protection against contact with hot parts or moving parts is provided by the enclosure, or by guards and interlocks.

To prevent the apparatus from being the initiator of a fire, construction standards specify careful selection of materials with respect to temperature rises of parts, minimum clearances between conductive parts to prevent short circuiting, and enclosure to prevent any arcs or sparks from leaving the equipment.

Instructions, warnings, and installation diagrams are assessed by third party approval authorities, as part of the approval process, in addition to evaluating conformity to construction rules. The user must install and use apparatus in accordance with these specifications and documents to ensure safety.

22.3 Equipment for Use Where Explosive Concentrations of Gas, Vapor, or Dust Might be Present

Equipment intended for use in hazardous locations is always marked for the hazardous locations in which use is permitted and the kind of protection incorporated, and it is almost always certified by an independent approval authority. The user may depend on this marking when selecting equipment for use.

22.3.1 Area Classification

Any hazardous area classification system defines the *kind* of flammable material that could be present, and the *probability* that it will be present. In North America and some other locations, two nomenclatures are in use to denote type and probability: Class, Group and Division, and the more recent international usage of Material Class and Zone. A third system, introduced in the European Community (EC), also addresses the nature and degree of hazard, and is applied also to locations where hazards other than an explosion exist. Class and Group or Material Group defines the nature of the hazardous material that may be present. Division or Zone indicates the probability of the location having a flammable concentration of the material.

Table 22-1: Material Classes and Groups and Divisions – National Electrical Code (NEC) Article 500

Class I **Gases and Vapors**	**Class II** **Combustible Dusts**	**Class III** **Ignitable Flyings**
Group A –Acetylene **Group B** – Hydrogen, butadiene, ethylene oxide, etc. **Group C** – Ethyl ether, cyclopropane, ethylene, etc. **Group D** – Gasoline, common solvents, natural gas, vinyl chloride. Propylene, styrene, etc.	**Group E** – Metallic dusts **Group F** – Coal dust, carbon black and similar carbonaceous dusts **Group G** – Grain dusts, plastics, sugar and other organic dusts	No Groups assigned. Typical materials are kapok, cotton linters, nylon, flax, wood chips, which are not normally in suspension
Criteria for grouping – Ease of ignition by electric arc and/or passage of explosion through narrow gap		
Division 1		
Continuous or intermittent hazard, as in below grade pits in Division 2 locations, near sources of release, such as packing glands, open kettles or vents, or where release of material and damage to electrical equipment may occur simultaneously.	Dust cloud of flammable concentration exists periodically, frequently, or intermittently, as near processing equipment. Where conducting dust may accumulate.	Where cotton, Spanish moss, etc. are manufactured or processed.
Division 2		
Areas below grade in unclassified location, or adjacent to Division 1 locations, or where flammable liquids are stored or piped in closed systems, or Div. 1 locations made nonhazardous by forced ventilation	Where failure of equipment may release a cloud or where dust layers deposited on floor or equipment may be resuspended as a cloud.	Areas where materials are stored or handled.

In international practice, mining hazards are denoted as Group I, due to potential presence of both methane and dust. Equipment for use in mines is constructed to protect against both hazards in a physically and chemically arduous environment. Industrial facilities are denoted as Group II and the gases and vapors are classified as follows.

Material group IIA - equivalent to Group D Material group IIB - equivalent to Group C Material group IIC - equivalent to Groups A and B	Alternatively: Group A = IIC Group B = IIB + Hydrogen Group C = IIB Group D = IIA

The degree of hazard is given by the Zone designation:

- Zone 0, Zone 20 – hazardous atmosphere or dust cloud may be present continuously or a large percentage of the time
- Zone 1, Zone 21 – hazardous atmosphere or dust cloud may be present intermittently
- Zone 2, Zone 22 – hazardous atmosphere or dust cloud present only abnormally, as after equipment or containment failure.

Thus, Class I, Division 1 includes Zone 0 and Zone 1, and Class I, Div. 2 = Zone 2.

And, Class II, Division 1 includes Zone 20 and Zone 21, and Class II, Div. 2 = Zone 22.

22.3.2 Temperature Class

Electrical apparatus often develops areas of sufficiently high temperature to ignite a material it may contact. To allow the user to ensure equipment will not become a thermal source of ignition, the manufacturer must indicate the value of the highest temperature reached by a part that is accessible to the explosive or combustible mixture.

At present this is most often done by specifying a Temperature Class. These are defined in the table below. Classes without a suffix are internationally recognized. Those with suffixes are defined in North America, because many intermediate temperature limits were defined in electrical codes or standards prior to 1971. As a practical matter, a T4 class is safe for all but carbon disulphide and perhaps one or two other vapors. The maximum surface temperature for some equipment for use in dusty locations may be controlled by the testing standard and may not be marked on the equipment.

Maximum Temperature		Temperature Class
°C	°F	
450	842	T1
300	572	T2
280	536	T2A
260	500	T2B
230	446	T2C
215	419	T2D
200	392	T3
180	356	T3A
165	329	T3B
160	320	T3C
135	275	T4
120	248	T4A
100	212	T5
85	185	T6

22.3.3 Selection of Apparatus

Once the classification of a location has been established, usually during plant design, one can select electrical apparatus which is safe to use in that location.

Figure 22-1 shows the types of protection against ignition of gases and vapors in common use, and their applicability. Ex d, Ex p, etc. are the International Electrotechnical Commission (IEC) designations for that type of protection, which are also recognized in the NEC (denoted AEx) and Canadian Electrical Code (denoted Ex or EEx) for marking apparatus for use in the several zones.

TYPE OF PROTECTION (See Note 1)											
Area Classification		Intrinsic safety		Explosionproof/ Flameproof	Pressurization (Note 2)		Type N Protection	Encapsulation		Increased Safety	Unprotected apparatus
					Isolate	Alarm					
		Ex ia	Ex ib	Ex d	Ex p	Ex p	Ex n	Ex ma	Ex mb	Ex e	
Zone 0	Division 1	Yes						Yes			
Zone 1		Yes	Yes	Yes	Yes (Note 3)	Ex n apparatus		Yes	Yes	Yes	
Zone 2/Division 2		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	If normally no ignition source
Unclassified		Yes	Yes	Yes	Yes	Yes	yes	Yes	Yes	Yes	Yes

Note 1. This figure illustrates the relationship among the types of protection and the Zone/Division notation. Only explosionproof, pressurized or intrinsically safe constructions are recognized for Class I, Division 1 use in North America. Few, if any, explosionproof and pressurized apparatus are installed in that small percentage of Division 1 locations that could be classified Zone 0. Types of Protection o (oil immersion) and q (powder filling) are Zone 1 techniques, but are not listed above because they are seldom used.

Note 2. "Pressurization" replaced the older term "purging" after it was recognized that flow more than that needed to assure a standardized minimum pressure inside an enclosure is unnecessary. "Purging" is now an initial deliberate flow through an enclosure to sweep out accumulated gases or vapors.

Note 3. In North American standards, a pressurization system which disconnects the power to ignition-capable apparatus in Division 1 when pressurization fails is Type X pressurization. In a Type Y system, loss of pressurization protecting apparatus suitable for Division 2 requires only an alarm. A Type Z system protecting ignition-capable equipment in Division 2 requires only an alarm. At present international standards recognize two constructions, with interlock and with alarm. Application is left to local codes of practice.

Figure 22-1: Applicability of Common Types of Protection

Table 22-2 describes the protection concept, salient construction features of each Type of Protection, and aspects of safe use specific to that Type of Protection. To ensure safe use of any Type of Protection, the user must install per instructions that accompany the apparatus—especially with regard to protection from the environment, shock, vibration, and, where specified, grounding and bonding. The user must inspect the installation frequently enough to ensure the integrity of the enclosure, and continuity of grounding and bonding, has not become impaired by accident or environmental attack.

Table 22-2. Protection Concept, Salient Construction Features, and User Duties Specific to Each Type of Protection

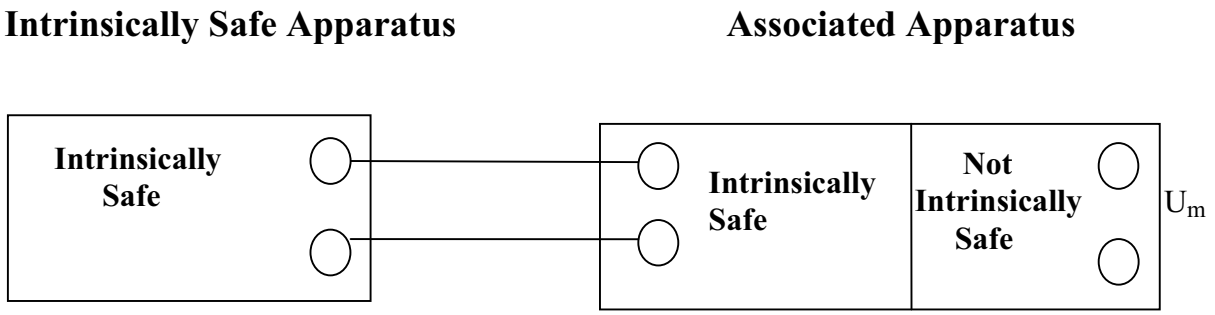
Type of Protection	Protection Concept	Salient Construction Features	User Duties to Maintain Safe Installation
d – flameproof/explosionproof	An explosion inside the enclosure is not transmitted to the outside flammable atmosphere through gaps, shaft clearances, of by damage to the enclosure.	Heavy walls, usually cast metal, but may be plastic in small sizes. Joints are tight fitting bolted flanges with narrow gaps or close fitting multiple threads. Shaft clearances very small. Small thick windows in reinforced mounting. Seals in wiring system complete the flameproof enclosure.	Ensure all bolts and threaded joints are tight and remain so. Ensure damage by corrosion or accident detected and corrected. Remove covers only if area non-hazardous, and ensure joints are clean, and undamaged before covers are replaced. Ensure all seals are installed per codes.
p – pressurized	Positive pressure of air, or sometimes inert gas, inside the enclosure prevents entry of flammable gases, vapors and dusts.	Electrical equipment may be of standard design, though cost of supplying leaking pressurization medium may require sealing of openings. User may supply controls, interlocks, alarms, and warning labels required to ensure safe operation, but certified assemblies of apparatus and controls are available.	Periodic check to ensure that pressure or flow-actuated alarm or interlock works as designed. Enforce rules against energizing without ensuring enclosure is free of explosive materials if purging is not automatic.
i – intrinsic safety	Extremely conservative construction rules and limits on current and voltage available, even after failures, reduce probability to nearly zero that any failure in component or wiring has enough energy to ignite a flammable material. The only standard protection concept recognized safe in Zone 0. Only protection concept generally suited for some servicing while energized without 'hot permit.' ia – safe in Zone 0 or 1. ib – safe in Zone 1.	Useful primarily for low power measurement and control applications (Note 1). Control of power ratings on components and stringent rules for spacings between conductors demands design for intrinsic safety from inception for apparatus located in Div. 1 or Zone 0/1 location. Isolating assemblies (barriers) can permit connection of general purpose equipment in a safe or Div. 2/Zone 2 location to intrinsically safe apparatus in the field. Stringent limitations on capacitance and inductance of field wiring and apparatus. .	Essential that installation instructions, especially limits of inductance and capacitance of field apparatus and wiring, and grounding, be observed. Administrative controls must ensure that changes to the installation or addition of a new device do not violate installation rules and restrictions, and that i(b) apparatus is used only in Zone 1, not Zone 0.

e – increased safety	Originated as conservative design to reduce probability of ignition-capable arc developing in non-sparking rotating machinery, but same principles applied to terminals, connection facilities, resistance heaters, transformers, etc.. Type e construction often used for connection facilities of flameproof enclosures, and in conjunction with flameproof enclosures in switchgear.	In rotating machinery stator-rotor gaps are wider than normal to avoid frictional sparking. Coils are conservatively designed to reduce temperature rise and must be impregnated. Conservative rules for stating Temperature Class and selecting overcurrent protection devices to limiting temperature rise during under short circuit and stalled rotor conditions. Internal connections are robust, soldered, welded, etc., and external connection terminals must not loosen under vibration, shock or temperature cycling and must maintain a low resistance connection.	Careful attention to specified ratings of protective devices, installation of external wiring. Safety is especially dependent on equipment being operated within all specified ratings.
m - encapsulation	Ignition capable elements sealed in encapsulant or potted to prevent exposure to flammable mixture. ma – safe after two faults of encapsulated components, no switching contacts permitted, more robust construction than mb. mb – safe after one fault. Less restrictive construction requirements.	Devices up to volume of 100cc encapsulated with specified minimum wall thickness except for volumes less than 1cc. Protection shall be maintained after fault(s). Creepage and clearances are defined which are presumed not to fault, as are certain components used at less than 2/3 rating. Encapsulant is tested for water absorption and dielectric strength. Device tested over rated temperature range to ensure temperature rating of encapsulant and surface temperature class are not exceeded.	Primary duty of user, in addition to ensuring a device is used only in zone permitted, is to ensure that encapsulated device is housed in a protective enclosure if it is not rated for unprotected mounting, and that ambient limits and power ratings are not exceeded in use.

n – Apparatus which is nonsparking in normal operation, -Normally sparking apparatus prevented from being an ignition source by one of several methods. - Circuits in which opens, shorts, or arcing contacts are energy limited in normal operation ao ignition does not occur.	nA - nonsparking apparatus. nC - sparking apparatus, isolated from flammable atmosphere by enclosed break device, hermetic sealing, sealing, non-incendive component, or encapsulation. nR - restricted breathing enclosure. nL - energy limited apparatus.	nA – rotating machines, low power measuring and control apparatus. nC –enclosed break device is low volume enclosure strong and tight so explosion doesn't propagate outside if it should occur. - hermetic seal is made by soldering, welding, etc. - sealed device, a small enclosure made tight by means other than fusing metal or glass. - nonincendive component, construction such that arcing is incapable of causing ignition. - restricted breathing enclosure may be large and house ignition-capable contacts, but is so tight that outside flammable atmosphere in Zone 2 will not result in inside atmosphere becoming flammable. Internal temperature rise must be low. - energy limited apparatus is assessed, in normal operation only, using ignition criteria for intrinsic safety.	User must be especially aware of the conditions and limitations of use required by the certification documents or instructions, as ambient conditions are very important in assessing safety of each design. Techniques depending on maintaining nonflammable atmosphere inside enclosure demand care in use, and inspection often enough to reveal any degradation of the enclosure integrity.
Note 1. Single limiting values for voltage, power or energy cannot be stated, but most systems and devices operate at less than 30V, and a few watts, depending on the explosive material classification. However, electrostatic paint spray systems operating at kilovolts have been approved as intrinsically safe. Typical ignition energies in the standard test apparatus are 40 microjoules or higher.			

All the protection techniques referred to in Figure 22-1 and Table 22-2, except intrinsic safety and energy limited Type n or Division 2 constructions, are device oriented. In principle, a user purchases the equipment, opens the box, and installs in accordance with the installation instructions and the applicable installation code—the NEC or CEC in North America. Intrinsic safety and energy limited design for Div. 2/Zone 2 applications are *system* oriented. Figure 22-2 illustrates an intrinsically safe system. A discussion of an energy limited system for Division 2/Zone 2 would be based on a similar diagram.

Intrinsically safe apparatus is composed entirely of intrinsically safe circuits. It can be described in terms of the maximum voltage, maximum current, and maximum power which may be impressed on its terminals; and, by the effective capacitance, inductance, or optionally, inductance to resistance (L/R) ratio, which can be seen looking into the terminals. Figure 22-2 shows only a single pair of terminals. There may be multiple pairs of terminals, each of which must be characterized by the characteristic values noted. The first symbol when two are shown is common in international standards. The second is common in North America, though both are permitted. Associated apparatus contains circuits that are intrinsically safe and circuits that are not intrinsically safe. If the nonintrinsically safe circuits are protected by some other technique, the apparatus may be located in a hazardous location. Otherwise, it must be located in an unclassified location. The intrinsically safe terminals are characterized by the maximum voltage, maximum current, and maximum power which can be delivered at the terminals, and the maximum values of capacitance, inductance, and optionally, L/R ratio, which may be connected safely to the terminals. Nonintrinsically safe terminals are characterized by the maxi-



U_i or V_{max} – max. safe applied voltage
I_i or I_{max} – max. safe input current
P_i or P_{max} – max. power input

C_i – total equivalent capacitance seen at input terminals
L_i – total equivalent inductance seen at input terminals
L_i/R_i – effective ratio of inductance to resistance at input terminals

U_o or V_{oc} – max. open circuit voltage
I_o or I_{sc} – max. output current
P_o – max. deliverable power

C_a or C_o – maximum permitted connected capacitance
L_a or L_o – maximum permitted connected inductance
L_a/R_a or L_o/R_o – maximum permitted ratio of inductance to resistance in connected circuit

Figure 22-2: Elements and Parameters of an Intrinsically Safe System

mum rated voltage that may be applied to them. This voltage, U_m, in Figure 22-2, is usually 250V dc or rms for equipment for connection to the power line, but it could be 24V or other low voltage for other equipment.

In the former case, the design will be based on some minimum prospective current available from the power line. In the latter case, the assessment is carried out assuming presence of the low voltage, but the certificate or control drawing may demand that the 24V be supplied from a protective transformer, constructed in accordance with the provisions of the standard. The intent of this mandate is to ensure that the transformer, which is not a part of the certified apparatus, is of a quality of construction that will reduce to an acceptably low value the probability that its failure allows a voltage higher than 24V to appear at the terminals. In countries following the IEC pattern, apparatus is usually certified as an entity, without regard to the specific design of the other apparatus to which it is connected in a system. Intrinsically safe apparatus is defined by the parameters indicated in Figure 22-2. In principle, the use of these parameters to select devices to be used in a system is straightforward if the intrinsically safe device is a two-terminal device. It is only necessary to ensure that U_o and I_o are equal to or less than U_i and I_i; and, that L_i, and C_i, are equal to or smaller than L_o and C_o. If the intrinsically safe apparatus is a two-wire device, and both wires are isolated from ground, an associated apparatus (barrier) must be installed in each wire. In North America it has now become a requirement on the manufacturer of the intrinsically safe or associated apparatus to provide a "control drawing" which provides the details of the permissible interconnections and any special installation requirements peculiar to that apparatus. This control drawing is assessed and verified by the certifying agency as part of its examination of the product.

Standards following the IEC pattern define two levels of protection of intrinsic safety: level of protection ia apparatus and level of protection ib apparatus.

Level of protection ia apparatus, suitable for Zone 0, will not cause ignition when the maximum permitted values are applied to its terminals:

- in normal operation with application of those non-countable faults which give the most onerous condition;
- in normal operation with application of one countable fault and those non-countable faults which give the most onerous condition; and
- in normal operation with application of two countable faults and those non-countable faults which give the most onerous condition.

Normal operation means that the apparatus conforms to the design specification supplied by the manufacturer, and is used within electrical, mechanical, and environmental limits specified by the manufacturer. Normal operation also includes open circuiting, shorting, and grounding of external wiring at connection facilities.

When assessing or testing for spark ignition, the safety factors to be applied to voltage or current are 1.5 in conditions a and b, and 1.0 in condition c. (These factors should properly be called "test factors." The real safety of intrinsic safety is inherent in the use of the sensitive IEC apparatus to attempt to ignite the most easily ignitable mixture of the test gas with hundreds of sparks. This combination of conditions is many times more onerous than any likely to occur in practice.)

North American Intrinsic Safety design standards are equivalent to ia intrinsic safety.

Level of protection ib apparatus, suitable for Zone 1, is assessed or tested under the conditions of a and b above, with a safety factor on voltage or current of 1.5 in the condition of a and b.

It is likely that Level of protection ic apparatus, equivalent to and replacing Type of Protection nL, suitable for use in Zone 2, will be standardized.

Figure 22-3 shows typical grounded and ungrounded two wire intrinsically safe circuits.

Figure 22-3 illustrates the principle that every ungrounded conductor entering the Division 1/Zone 0, or 1 location in this case where the transmitter or transducer is located, must be protected against unsafe voltage and current by appropriate associated apparatus. The boxes with three terminals represent barriers, independently certified protective assemblies, certified and rated according to the national standard. Nonintrinsically safe devices connected to the barrier need only be suitable for their location, and must not contain voltages higher than the Um rating of the barrier.

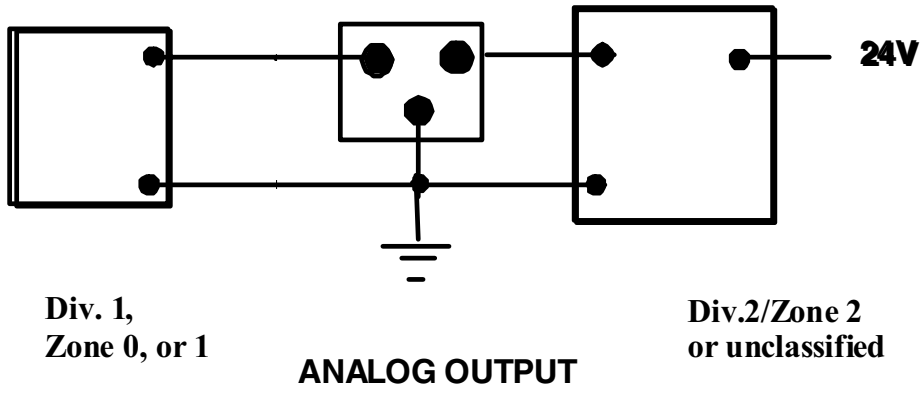
Many barriers are passive, consisting of current limiting resistors and voltage limiting diodes in appropriate configuration and redundancy to meet the requirements of the standard. Others have active current or voltage limiting. Both types may be combined with other circuitry for regulating voltages, processing signals, etc. The user should follow the recommendation of the intrinsically safe apparatus manufacturer in the control drawing, or discuss the selection of appropriate barriers with the barrier vendor, all of whom have proven configurations for many field mounted devices.

22.4 Equipment for Use in Locations Where Combustible Dust May be Present

Table 22-3 illustrates the methods approved in 2005 for use in Class II locations in the NEC.

Definition of Zones 20, 21, and 22 in the Canadian Electrical Code is pending.

Dust ignitionproof construction is designed and tested to ensure no entry of dust under swirling test conditions while cycling equipment to create a pressure drop to draw dust into the enclosure. Temperature rise of the enclosure surface is determined after a thick layer of dust has accumulated on the enclosure. Dust tight construction may or may not be tested by a third party. If so, the test is essentially the same, but the equipment is cycled only once during the dust test.



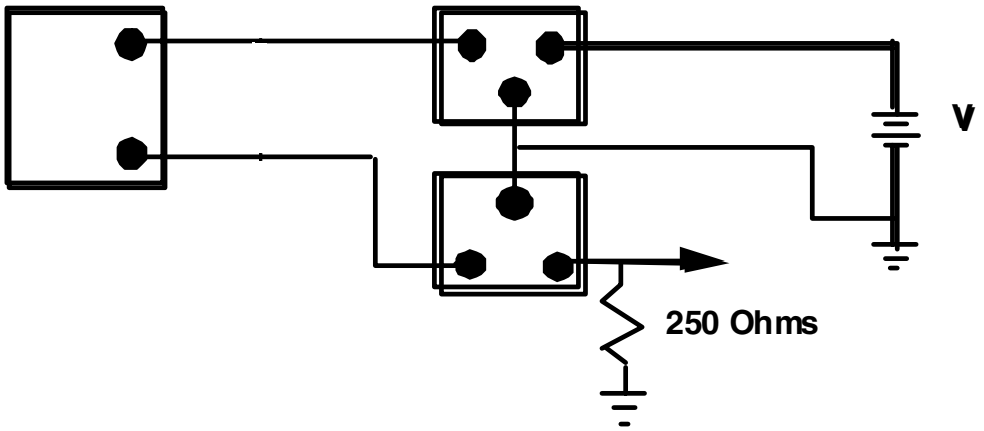


Figure 22-3: Transmitter Input and Controller Output

Table 22-3: Methods of Protection for Equipment in NEC Class II Locations

Type of Protection	Division 1		Division 2
	Zone 20	Zone 21	Zone 22
Dust ignitionproof	No	Yes	Yes
Intrinsic Safety	Yes	Yes	Yes
Pressurized	No	Yes	Yes
Dusttight	No	No	Yes
Nonincendive circuit	No	No	Yes
Nonincendive equipment	No	No	Yes

In European practice the user has responsibility for determining that equipment surface temperature under the expected accumulation of dust is safely below the ignition temperature of the dust involved. Equipment enclosure standards reflect this difference. The IEC standard for enclosures originally standardized constructions of both practices. Practice A is similar to American dust ignitionproof and dust tight enclosure. Practice B is European, in which enclosures for Zone 21 are tested under vacuum to be dust tight, i.e., degree of protection IP6X. Enclosures for Zone 22 are permitted to have entry of

some dust, but not enough to impair function or decrease safety. However, it is expected that, in the near future, there will be greater harmonization of practice.

The recommendations adopted by IEC in 2004 are summarized in Table 22-4.

Table 22-4: IEC Recommendations

Type of Dust	Zone 20	Zone 21	Zone 22
Nonconductive	tDA20 tDB20 iaD maD	tDA20 or tDA21 tDB20 or tDB21 iaD or ibD maD or mbD pD	tDA20 or tDA21or tDA22 tDB20 or tDB21 ortDB22 iaD or ibD maD or mbD pD
Conductive	tDA20 tDB20 iaD maD	tDA20 or tDA21 tDB20 or tDB21 iaD or ibD maD or mbD pD	tDA20 or tDA21or tDA22 IP6X tDB20 or tDB21 ortDB22 iaD or ibD maD or mbD pD

The suffix D after the symbol for Type of Protection indicates the version of that technique intended for use with dusts, sometimes with reduced construction and test requirements. The symbol "t" refers to Protection by Enclosure, essentially dust tight or dust ignitionproof construction, and suffixes A and B are as discussed above. It is likely that IEC and U.S. standards for area classification and equipment selection in dusty areas will grow in closer agreement in the coming years.

22.4.1 The Label Tells about the Device

In North America, apparatus for use in Division 1 or 2 is marked with the Class and Group for which it is approved. It may be marked Division 1, but must be marked Division 2 if suitable for Div. 2 only. The temperature code, discussed below, is also shown.

Examples: Class I, Groups A-C, Division 1 T4

Class I, Groups C, D, Division 2 T6

In addition equipment approved for Class I, Division 1 or Division 2 may also be marked with Class, Zone, Gas Group, and temperature code. For the above examples the additional marking would be:

Class I, Zone 1, IIC, T4

Class I. Zone 2, IIA, T6

In the United States, apparatus approved for use specifically for use in Zone 0, Zone 1, or Zone 2 shall be marked with Class, Zone, AEx, the symbol for the method of protection, the applicable gas group, and temperature code.

Examples: Class I, Zone 0 AEx ia IIC, T4

Class I, Zone 1 AEx m IIC, T6

In Canada, the Class and Zone markings are optional and AEx is replaced by Ex or EEx, which are the symbols for explosion-protected apparatus conforming to IEC and CENELEC standards, respectively. In countries using only the zone classification system only marking such as those below are required.

Ex ia IIC, T4

EEx ia IIC, T4

If more than one Type of Protection is used, all symbols are shown: i.e., Ex d e mb IIC, T4.

In addition to the markings specifically relevant to explosion protection, a label will usually also provide additional information such as:

- Name and address of manufacturer
- Degree of protection afforded by the enclosure, NEMA or CSA enclosure rating in North America, or the IP code in countries following IEC patterns
- Symbol of certifying authority and an approval document number
- Voltage, current and power ratings
- Pressure ratings, if applicable.

If equipment is certified by a recognized approval agency, the user may be assured it is safe if it is installed and used according to the instructions supplied by the manufacturer. Conditions of use and installation instructions may be incorporated by reference in the certification documents or in drawings provided by the manufacturer. In all cases, it is essential the user install equipment in accordance with local codes, and operate it within its electrical supply and load specifications and rated ambient conditions.

22.5 For More Information

Few users need more detail about the fire and shock hazard protection that's built into the equipment they buy, but those who wish to dig deeper may consult the documents produced by the standards committees ISA-SP82, Electrical and Electronic Instrumentation, and IEC Technical Committee 66. ISA-SP82 developed ANSI/ISA-82.02.01-2004 (IEC 61010-1 Mod) - *Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use*, and the IEC group produced IEC 61010 - *Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use*. A broader understanding of area classification and the types of protection does, however, aid safe application of equipment in classified locations.

Publications by ISA-SP12, Electrical Equipment for Hazardous Locations, include ANSI/ISA-60079-0 (12.00.01)-2005 - *Electrical Apparatus for Use in Class I, Zones 0, 1 & 2 Hazardous (Classified) Locations*. That series includes information about design and use of electrical apparatus for classified locations, as do the U.S. versions of the IEC standards for the Types of Protection discussed above.

IEC publications form the basis of an increasing number of national standards of IEC member countries, the local versions of which can be purchased from the national member body of IEC.

The following is a list of standards addressing use and application of electrical apparatus rather than design:

ISA

ANSI/ISA-12.01.01-1999 - *Definitions and Information Pertaining to Electrical Instruments in Hazardous (Classified) Locations*

ISA-TR12.2-1995 - *Intrinsically Safe System Assessment Using the Entity Concept*

ISA-RP12.2.02-1996 - *Recommendations for the Preparation, Content, and Organization of Intrinsic Safety Control Drawings*

ANSI/ISA-TR12.24.01-1998 (IEC 60079-10 Mod) - *Recommended Practice for Classification of Locations for Electrical Installations Classified as Class I, Zone 0, Zone 1, or Zone 2*

IEC

IEC 60079-10 *Electrical apparatus for explosive gas atmospheres - Part 10: Classification of hazardous areas*

IEC 60079-17 *Electrical apparatus for explosive gas atmospheres - Part 17: Inspection and maintenance of electrical installations in hazardous areas (other than mines)*

IEC 61285 *Industrial-process control - Safety of analyser houses*

IEC 61779-6 *Electrical apparatus for the detection and measurement of flammable gases - Part 6: Guide for the selection, installation, use and maintenance of apparatus for the detection and measurement of flammable gases*

NFPA

NFPA 70 *National Electrical Code®*

NFPA 70B *Recommended Practice for Electrical Equipment Maintenance*

NFPA 70E *Standard for Electrical Safety in the Workplace*

NFPA 496 *Standard for Purged and Pressurized Enclosures for Electrical Equipment*

NFPA 497 *Recommended Practice for the Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas*

NFPA 499 *Recommended Practice for the Classification of Combustible Dusts and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas*

The websites of the above-mentioned organizations are found at:

www.isa.org
www.iec.ch
www.nfpa.org

Many manufacturers provide free literature, often viewable on their Websites, which discuss the subjects of this article.

For an in-depth treatment of the science behind the types of protection and the history of their development, as well as design, installation, and inspection of installations one can consult the book *Electrical Instruments in Hazardous Locations*, published by ISA.

For discussion of installation of apparatus see:

Schram, P.J. and M. W. Earley. *Electrical Installations in Hazardous Locations*. NFPA, 1998.

McMillan, Alan. *Electrical Installations in Hazardous Areas.* Butterworth-Heinemann, 1998.

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About the Author

Ernie Magison was active in standards development in ISA, IEC, and NFPA for four decades. He has authored 40 articles, as well as papers and several books, most focusing on application of electrical apparatus in potentially explosive atmospheres. He has taught many short courses and consulted in the field.